Finite regular covers from the homology of an end-coset complex

Human Compression project

10 October 2026 — author edition for independent review

1. The finite question and the scope of this paper

Let $G = \langle r_0, \dots, r_{n-1} \rangle$, with $n \geq 3$, be a finite string group generated by distinct involutions. Put

$$A = \langle r_0, \dots, r_{n-2} \rangle, \qquad B = \langle r_1, \dots, r_{n-1} \rangle, \qquad C = \langle r_1, \dots, r_{n-2} \rangle, \qquad D = A \cap B.$$

Throughout this paper, the marked groups A and B are string C-groups. The only possible failure of the intersection property in G is therefore $D \neq C$. We retain the standard recursive intersection criterion: a string group whose two end groups are C-groups is a C-group precisely when their intersection is the middle group.

A marked epimorphism $H \to G$ is *locally faithful* here if it restricts to isomorphisms on both end groups. This term refers to these two specified subgroups. Our construction decides, by a finite integer matrix, whether there is such a C-cover with finite abelian kernel. It also gives an exact test for a kernel of elementary abelian prime-power order. The hypothesis of local faithfulness is essential.

The unrestricted minimal-cover theorem established in the retained papers `phase216_general_covers.md` and its independent review `phase222_general_cover_review.md` allows infinite regular covers. It does not settle whether every finite polytope with non-C monodromy has at least two finite minimal regular covers. Nothing in the present paper settles that question in full. The general result proved here is a conditional finite construction and an exact obstruction to one restricted construction method.

For a genuine example, the Tomotope has a locally faithful regular cover of order 73728, with kernel C_2^2. Nevertheless it has no locally faithful cover with a nontrivial finite p-group kernel for any odd prime p. The latter statement is compatible with the already known infinitely many finite minimal covers of the Tomotope: such odd-primary covers must enlarge at least one end group. Monson–Pellicer–Williams already construct two covers of order 73728 in Examples 4.1–4.2, a minimal cover of order 36864 in Proposition 4.5, and infinitely many minimal covers in Theorem 5.9. None of those existence conclusions is a new conjecture solution here. Equality of an order does not identify our marked cover with either earlier example: their end-group data must also be compared.

2. The universal completion and its kernel

Form the amalgam

$$V = A *_C B$$

and identify the common marked generators $r_1, \dots, r_{n-2}$. The maps of the factors into G induce an epimorphism $\pi : V \to G$. Its restriction to each factor is injective. Let

$$U = V/\langle\langle (r_0 r_{n-1})^2 \rangle\rangle_V, \qquad \Pi = \ker(U \to G).$$

Every marked locally faithful cover of G, without an assumption on its kernel, is a quotient of U. Indeed its two end groups realize the two specified factors, their common generators realize C, and the distant generators commute. Conversely a quotient of U through which $U \to G$ factors remains faithful on each factor: an element of a factor killed in that quotient would already be killed in G.

We use the usual Bass–Serre tree T of the amalgam. Its vertices are $V/A \sqcup V/B$, its edges are V/C, and vC joins vA to vB. The subgroup $F = \ker \pi$ intersects every conjugate of a factor trivially. Consequently it acts freely on T, and

$$\Gamma = F\backslash T$$

is the finite graph with vertices $G/A \sqcup G/B$ and edges G/C. In particular $F \cong \pi_1(\Gamma)$, after a base point has been chosen. This is the ordinary free-action theorem for a group acting on a tree; no residual-finiteness assertion is involved.

Orient every edge from its A-vertex to its B-vertex, and denote the edge gC by e_g. For each $g \in G$, attach a square with cellular boundary

$$z_g = e_g - e_{gr_0} + e_{gr_0 r_{n-1}} - e_{gr_{n-1}}. \tag{2.1}$$

These are closed edge paths. The first two edges have the same A-endpoint, the middle two the same B-endpoint, and the other two endpoint equalities follow from the commutation of r_0 and r_{n-1}. Repeated or degenerate attaching paths cause no difficulty. We retain every translated square, so no orbit-selection convention is required. Call the resulting complex X. The implementation uses right cosets instead: inversion sends gC to Cg^{-1} and carries (2.1) to its right-coset version. It preserves the graph, the full translated family of square relations and the set of parallel-edge defects.

Proposition 2.1. There is a natural isomorphism $\Pi \cong \pi_1(X)$, up to the usual choice of base paths.

Proof. The relator $w = (r_0 r_{n-1})^2$ lies in F. Since F is normal and $V/F = G$, the normal closure of w in V is the normal closure in F of one conjugate of w for each element of G. In the graph $F\backslash T$, the loop for each such conjugate is a base-path conjugate of the corresponding square (2.1), with a possible reversal of orientation. This can also be seen directly by following the alternating factor word $r_0 r_{n-1} r_0 r_{n-1}$ through the tree. Attaching all the squares therefore divides F by exactly this normal closure. The resulting group is the kernel of $U \to G$, as required. □

Both factors remain embedded in U, since the map to G is injective on them. The proposition does not assert that U is residually finite, finite, or a C-group.

3. Which homology classes repair the intersection?

For each $g \in D$, the edges e_g and e_1 have the same two endpoints. Thus

$$\delta_g = [e_g - e_1] \in H_1(X; \mathbb{Z}) \tag{3.1}$$

is defined. It is zero for $g \in C$. Different right cosets gC in D give the complete finite list to be checked.

For an integer $m \geq 2$, define

$$L_m = \ker(\Pi \longrightarrow H_1(X;\mathbb{Z})/mH_1(X;\mathbb{Z})), \qquad H_m = U/L_m. \tag{3.2}$$

The subgroup L_m is characteristic in Π, hence normal in U. The marked group H_m is finite, with

$$|H_m| = |G|\,|H_1(X;\mathbb{Z})/mH_1(X;\mathbb{Z})|.$$

For a prime p, the kernel is elementary abelian, of dimension $\dim_{\mathbb{F}_p} H_1(X;\mathbb{F}_p)$. The equality with mod-$p$ homology follows either from the cellular complex or from the universal coefficient theorem, since $H_0(X;\mathbb{Z})$ is torsion-free.

Theorem 3.1 (exact abelian cover criterion). The group H_m is a string C-group if and only if the image of δ_g in $H_1(X;\mathbb{Z})/mH_1(X;\mathbb{Z})$ is nonzero for every $g \in D \setminus C$.

Proof. The two end groups of H_m are exactly the embedded copies of A and B, and are therefore C-groups. Fix $g \in D$. Let a_g and b_g be its unique lifts in the two factors. Their equality in H_m is equivalent to the triviality of $a_g b_g^{-1}$ in the kernel of $H_m \to G$.

The tree path from A to $a_g b_g^{-1} A$ traverses the edge $a_g C$ positively and the edge $a_g b_g^{-1} C$ negatively. In $F\backslash T$, these become e_g and e_1. Under Proposition 2.1 and abelianization, the element $a_g b_g^{-1}$ therefore represents exactly δ_g. Thus the two lifts agree if and only if the stated homology class vanishes.

Every element in the intersection of the two lifted end groups projects to D; since both restrictions are injective, there is at most one lift in each factor over a given element. For $g \in C$, these lifts are already the same element of the common middle group. Hence the lifted intersection equals that middle group exactly when no further g has equal lifts. The recursive intersection criterion now proves both directions. In particular, the argument controls the complete intersection, not merely its projected image. □

Theorem 3.2 (completeness for abelian kernels). A finite locally faithful C-cover of G with abelian kernel exists if and only if every δ_g, $g \in D \setminus C$, is nonzero in $H_1(X;\mathbb{Z})$. For a fixed prime p, a locally faithful C-cover with elementary abelian p-kernel exists if and only if H_p is a C-group.

Proof. Any locally faithful cover is a quotient of U. If its kernel is abelian, the induced epimorphism from Π to that kernel factors through the abelianization $H_1(X;\mathbb{Z})$. A vanishing δ_g cannot be separated in that quotient. If the kernel is elementary abelian of exponent p, this factorization further passes through $H_1(X;\mathbb{Z})/pH_1(X;\mathbb{Z})$.

Conversely, a finitely generated abelian group has a finite quotient of the form M/mM separating any specified finite collection of nonzero elements. To see this, write $M = \mathbb{Z}^b \oplus T$, take m divisible by the exponent of T, and make m larger than all absolute values of the nonzero coordinates needed to distinguish the finitely many elements with a nonzero free part. Apply this observation to the classes δ_g, and then apply Theorem 3.1. The fixed-prime assertion follows from the same theorem and the preceding factorization. □

Corollary 3.3 (a prime-kernel obstruction). If $H_1(X;\mathbb{F}_p) = 0$, no locally faithful cover of G has a nontrivial finite p-group kernel.

Proof. A nontrivial finite p-group has a quotient of order p. A surjection from Π to such a kernel would therefore give a nonzero homomorphism $\Pi \to \mathbb{F}_p$, contrary to the hypothesis. □

4. A conditional finite-multiplicity theorem

Theorem 4.1. Suppose G is not a C-group. If Theorem 3.1 succeeds for two distinct primes p, q, then G has at least two finite minimal C-covers. If all the classes δ_g, for $g \in D \setminus C$, are nonzero in $H_1(X; \mathbb{Q})$, then it has infinitely many finite minimal C-covers.

Proof. A finite C-cover has a minimal C-quotient over G, because it has only finitely many normal subgroups. Any such minimal quotient of H_p has kernel over G of p-power order. A common minimal quotient of H_p and H_q would have kernel order dividing relatively prime powers and would therefore equal G, which is not a C-group. Thus the minimal descendants are distinct.

For the second assertion, decompose the finitely generated group $H_1(X; \mathbb{Z})$ as in Theorem 3.2. A class with nonzero free part survives modulo every prime outside the finite set dividing all its free coordinates. There are finitely many classes to consider. Consequently all but finitely many primes satisfy Theorem 3.1, and the preceding coprimality argument separates their minimal descendants. □

This condition is sufficient, not necessary. In particular it fails for the Tomotope below, despite the known infinite multiplicity there.

5. Exact finite examples and the boundary of the method

The input for the Tomotope is the literal flag-adjacency model reconstructed from Monson–Pellicer–Williams in the retained, independently reviewed surface-cover package. Its 192 flags generate a monodromy group of order 18432. We retain that primary-to-model bridge; the present computations operate on its actual four permutations.

The three tested markings give the following exact data. The first two are group diagnostics; this paper does not claim that the marking of order 32 or the marking named D256 has independently established polytopal monodromy status.

Input	Base order	End, middle, overlap orders	Vertices, edges	Integral homology	Mod-2 cover order
Diagnostic 32	32	$16, 16, 4, 8$	$4, 8$	C_2	64
D256	256	$64, 64, 8, 16$	$8, 32$	C_2^2	1024
Tomotope	18432	$576, 1152, 24, 48$	$48, 768$	C_2^2	73728

For the Tomotope, an independent integer computation chooses a maximal tree, removes its edges from the cycle coordinates, and obtains a presentation with 721 generators. Integral unit elimination removes 719 of them. The remaining relations generate exactly $2\mathbb{Z}^2$, and the unique nontrivial parallel-edge class has nonzero image modulo 2. This proves the displayed integral homology group, rather than inferring it from computations in a few characteristics. The independent checker and its complete operation certificate are retained from phase245.

A separate construction realizes H_2 as four explicit permutations on 73728 points. GAP verifies that their group acts regularly, that the generator-bound map to the literal Tomotope monodromy is an epimorphism with elementary abelian kernel of order 4, that the two end groups have orders 576 and 1152, and that all 256 pairs of standard parabolics have the required intersection. This is a direct finite check of the cover, separate from the topological existence argument. By the universal

property and the elementary abelian kernel, this marked group is a quotient of H_2; their equal orders show that it is exactly H_2.

Proposition 5.1. Up to isomorphism of marked covers over the Tomotope monodromy, there are exactly two locally faithful C-covers with abelian kernel: the group of order 73728 constructed above and its unique C-quotient of order 36864. The latter is the previously constructed cover of Monson–Pellicer–Williams, Proposition 4.5.

Proof. The integral homology calculation and Theorem 3.2 show that every such cover is a quotient of H_2. Its kernel over the base is C_2^2. Exact enumeration in the native permutation group finds precisely one subgroup of order 2 normal in H_2. The actual lift discrepancy lies outside this subgroup. Thus the corresponding quotient remains a C-group by Theorem 3.1, whereas quotienting by the whole kernel returns the non-C group G. There are no other normal subgroups in the kernel, so these are all the possibilities. The order-36864 quotient is minimal because its kernel over G has prime order. The cover in the cited Proposition 4.5 has exactly these end-group orders and an epimorphism to G with kernel of order 2. It is therefore locally faithful, has abelian kernel, and must be this unique quotient. □

Since $H_1(X;\mathbb{Z}) = C_2^2$, Corollary 3.3 applies to every odd prime. Thus an odd-primary regular cover of the Tomotope cannot preserve both of these end groups. The surface construction in the retained phase211 paper does enlarge the ends. There is no contradiction between the two constructions.

6. The kernel-containment issue

Write W for the universal string group, and K for the kernel of its map to G. A regular cover is specified by a normal subgroup $N \leq K$ such that W/N is a C-group. A quotient of this cover corresponds to a larger normal subgroup. Consequently, if one starts from an admissible normal subgroup N_0 and tries to make its cover finite by taking a quotient, one needs

$$N_0 \leq N \leq K$$

while retaining the intersection property.

The finite-cover theorem of Monson–Schulte proves that every finite string group has a finite C-cover. Applied to a finite quotient W/L, it supplies a subgroup $M \leq L$, because the new C-group maps *onto* that quotient. It does not supply $L \leq M$, and a relator imposed in L may disappear in the cover. This is why finite-cover existence does not directly make the arbitrary-cover proof finite.

There is a useful separate sufficient condition. Let Q be a residually finite C-group mapping onto finite G, and suppose every proper standard parabolic of Q is finite. For every pair of proper index sets I, J, collect the nonidentity elements xy^{-1}, with $x \in Q_I$, $y \in Q_J$. The total collection is finite. Residual finiteness gives a finite-index normal subgroup avoiding it; intersect this subgroup with $\ker(Q \to G)$. In the resulting finite quotient, equal images of x and y force $x = y$. Thus proper parabolics embed and all their intersections are preserved. Intersections involving the full group are automatic, so the quotient remains a C-group over G.

This argument uses the finiteness of the proper parabolics in an essential way. The arbitrary-cover construction preserves infinite proper parabolics, and it cannot be inserted into this argument without an additional theorem. Nor does the presentation $U = A *_C B/\langle\langle (r_0 r_{n-1})^2 \rangle\rangle$ by itself establish that U is residually finite or a C-group.

7. Attribution and verification boundaries

The amalgam tree, its free kernel, cellular homology, and the structure theorem for finitely generated abelian groups are classical inputs. The recursive C-group intersection criterion and the correspondence between marked C-groups and regular abstract polytopes are retained standard theory. Monson–Schulte's finite-cover theorem is retained with the direction of its map stated explicitly. The Tomotope model and its known minimal-cover multiplicity are retained prior work, with the literal model bridge supplied by the already reviewed phase211/218 package.

The contribution of this paper is the explicit end-coset complex, its complete parallel-edge criterion for locally faithful abelian covers, the finite-multiplicity consequence under rational survival, and the exact Tomotope obstruction to all locally faithful odd-primary covers. No historical priority claim is made for these formulations. No whole-source replacement, principal-source count, compression rate, or newly solved general conjecture is claimed.

The topological identifications and uniform group arguments are written proofs. The cell matrices, integral Smith calculation, explicit voltage permutations and complete finite intersection checks are computational certificates. They are not a Lean formalization of the topology or of the general finite minimal-cover question. The accompanying Lean file proves the quotient intersection criterion for two subgroup images using actual homomorphisms, the preservation of end-group faithfulness under a quotient over the base, and the direction of kernel containment. It does not formalize the Bass–Serre identification, cellular homology, the finite matrix, or the minimal-cover application.